

5G Technology

Introduction

5G is the fifth generation of mobile communication offering higher speeds, lower latency, and massive device connectivity.

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Features

Enhanced Mobile Broadband, Ultra-Reliable Low Latency Communication, and Massive Machine Type Communication.

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Architecture

5G uses a flexible core network, small cells, beamforming, and Massive MIMO.

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Technologies

Network slicing, edge computing, virtualization, and millimeter-wave spectrum improve performance.

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Applications

Smart cities, autonomous vehicles, IoT, telemedicine, AR/VR, industrial automation, and cloud gaming.

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Advantages

High speed, low latency, greater capacity, energy efficiency, and improved user experience.

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Challenges

Deployment cost, spectrum availability, infrastructure upgrades, and security concerns.

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Future Scope

Supports AI, Industry 4.0, connected healthcare, and next-generation communication services.

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Viva Questions

What is 5G? Explain Massive MIMO. What is network slicing? Difference between 4G and 5G?
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